A Sharp Bound for Centers of Composite Planar Rotations

Research audit

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Abstract

An exact elementary counterexample shows that the center of a composition of positive planar rotations need not belong to the convex hull of the individual centers, even when their angle sum is at most π. We prove the sharp bound $(D/2)\tan(\Theta/4)$ for the distance of the composite center from that convex hull. The upper bound and sharpness are fully checked in Lean 4.32.2 with mathlib v4.32.2. Originality remains unestablished. An unfolding application still needs a quantitative separation margin; this result neither resolves Dürer's conjecture nor disproves a prismatoid unfolding theorem.

1 Source and scope

Proposition 9 of O'Rourke's *Prismatoid Band-Unfolding Revisited*, arXiv:2603.09813v3, states exact convex-hull containment for a composite rotation center when the total angle is at most π. Its Lemma 8 uses this ingredient for chain separation. The inspected HTML and PDF page 14 agree on the unrestricted wording. The latest listed version is dated 18 September 2026; no later correction was located in the bounded search. The author's separate hosted copy could not be retrieved reliably.

The author's earlier 2017 addendum explicitly distinguishes a small-angle weighted-center approximation from the true center, which can lie outside the hull. The exact rotation formulas themselves are classical; see, for example, Needham's *Visual Complex Analysis*, pp. 20–21. A failed ingredient does not refute the later main theorem: a valid chain-specific replacement may still prove the needed conclusion. This observation concerns only the stated planar-rotation ingredient.[1]

2 Exact counterexamples

Use positive angles counterclockwise, and write

$$R_{c,\theta}(z) = e^{\mathrm{i}\theta}z + (1 - e^{\mathrm{i}\theta})c.$$

Take $c_1 = (0, 0)$, $c_2 = (1, 0)$ and both angles $\pi/2$. Then

$$R_{c_1,\pi/2}(x, y) = (-y, x), \qquad R_{c_2,\pi/2}(u, v) = (1 - v, u - 1),$$

and hence

$$F(x, y) = R_{c_2,\pi/2}R_{c_1,\pi/2}(x, y) = (1 - x, -1 - y).$$

The equations $F(x, y) = (x, y)$ force $(x, y) = (1/2, -1/2)$. Every rotation fixes its center. Thus F cannot be a rotation about any point of the segment $[c_1, c_2]$. Both input angles are positive and

[1] Sources: `https://arxiv.org/abs/2603.09813v3` and `https://arxiv.org/abs/1709.02433`.

their sum is exactly π. Reversing composition order or reversing both angle signs reflects the center to $(1/2, 1/2)$, still outside the segment.

The endpoint π is not essential. Choose a first-quadrant angle θ with $\cos\theta = 3/5$ and $\sin\theta = 4/5$. Then $2\theta < \pi$, and the same two-center composition has unique center $(1/2, -1/4)$. More generally, two equal angles $0 < \theta < \pi$ give

$$c = \frac{c_1 + c_2}{2} - \frac{1}{2}\tan(\theta/2)\,\mathrm{i}(c_2 - c_1).$$

Thus exact hull containment fails even at arbitrarily small positive angles.

Formal verification boundary. The accompanying `RotationAudit.lean` defines ordinary real-plane rotations and proves the quarter-turn formula, angle conditions, unique fixed point, and impossibility of any rotation representation with center on the segment. It also proves the algebraic fixed-point statement for the rational matrix example. The reproducible build result is recorded in the separate check log. The rational example's strict angle interpretation is not currently part of that Lean certificate. The quarter-turn certificate successfully elaborated with pinned Lean 4.32.2 and mathlib v4.32.2; its axiom audit reports only `propext`, `Classical.choice` and `Quot.sound`. No polyhedral unfolding has been formalized.

3 The sharp diameter estimate

Theorem 1 (Sharp diameter estimate)**.** *Let $m \geq 1$ be finite, let $c_1, \ldots, c_m \in \mathbb{C}$, let $\omega_j \geq 0$ be angles in radians, and suppose $0 < \Theta = \sum_j \omega_j \leq \pi$. Set*

$$F = R_{c_m,\omega_m} \circ \cdots \circ R_{c_1,\omega_1}, \quad C = \mathrm{conv}\{c_1, \ldots, c_m\}, \quad D = \max_{j,k} |c_j - c_k|.$$

The unique center c of F satisfies

$$\mathrm{dist}(c, C) \leq \frac{D}{2}\tan(\Theta/4).$$

For each $\Theta \in (0, \pi]$, two equal-angle rotations about the endpoints of a segment attain equality.

Proof. Let $t_0 = 0$, $t_j = \sum_{k \leq j} \omega_k$, and $s_j = \Theta/2 - t_j$. Direct composition gives

$$c = \sum_{j=1}^{m} \lambda_j c_j, \qquad \lambda_j = \frac{e^{\mathrm{i}(\Theta - t_j)}(1 - e^{\mathrm{i}\omega_j})}{1 - e^{\mathrm{i}\Theta}}.$$

The denominator is nonzero. With $\lambda_j = p_j + \mathrm{i}q_j$,

$$p_j = \frac{\sin s_{j-1} - \sin s_j}{2\sin(\Theta/2)},$$
$$q_j = \frac{\cos s_j - \cos s_{j-1}}{2\sin(\Theta/2)}.$$

Since s_j decreases within $[-\pi/2, \pi/2]$, every $p_j \geq 0$. Telescoping yields $\sum p_j = 1$ and $\sum q_j = 0$. Therefore $w = \sum p_j c_j$ lies in C.

The sequence $\cos s_j$ increases and then decreases, including the possible single step that straddles zero. Its total positive increment is its maximum minus its initial value. Thus, with $Q = \sum_j \max(q_j, 0)$,

$$Q \leq \frac{1 - \cos(\Theta/2)}{2\sin(\Theta/2)} = \frac{1}{2}\tan(\Theta/4).$$

If $Q = 0$, then $c = w$. Otherwise the positive and negative coefficients define two convex combinations $u, v \in C$ such that $\sum q_j c_j = Q(u - v)$. Any two convex combinations of a set of diameter D are at distance at most D, by applying the triangle inequality to the double convex combination of their differences. Consequently

$$|c - w| = \left|\mathrm{i}\sum q_j c_j\right| = Q|u - v| \le QD \le \frac{D}{2}\tan(\Theta/4).$$

This proves the estimate. For equality, take $c_1 = 0$, $c_2 = D$ and $\omega_1 = \omega_2 = \Theta/2$. The center is $D/2 - \mathrm{i}(D/2)\tan(\Theta/4)$, whose distance from $[0, D]$ is the claimed bound. □

A usable margin condition. If a unit vector a satisfies $a \cdot c_j \le b - \delta$ for all j, then $a \cdot c < b$ whenever $\delta > (D/2)\tan(\Theta/4)$. Apply the estimate to the displayed convex combination w. An unfolding application must still prove this margin from its actual chain geometry.

4 Formal theorem and verification scope

The main theorem in `RotationCenterBound.lean` uses a finite list of actual rotations. It assumes nonnegative individual angles, $0 < \Theta \le \pi$, $D \ge 0$, and pairwise center distances at most D. It proves that the composite is a rotation with a unique fixed point c, and constructs a hull point w with $|c - w| \le (D/2)\tan(\Theta/4)$. No coefficient estimate or trigonometric claim remains as a hypothesis. Choosing the maximum pairwise distance gives the theorem above.

`RotationSharpness.lean` proves the matching lower bound for every point in the two-center hull and existence of an attaining hull point, for every permitted Θ and $D \ge 0$. The endpoint distance is certified as exactly D. The semantic fixture `RotationExpectedStatement.lean` rewrites the same composite using the literal map $e^{\mathrm{i}\theta}(z - c) + c$ and the ordinary list sum of angles.

The six-module closure uses only `propext`, `Classical.choice`, and `Quot.sound`. The accompanying source manifest and bounded replay script pin the checked sources and compiler assumptions. The formal proof bounds the positive cosine variation by a clipped monotone potential, avoiding an unformalized unimodality claim. The preceding half-plane consequence is a mathematical corollary, not an additional Lean declaration in this package.

5 Priority and remaining questions

The proof has also passed a separate mathematical audit. Bounded primary-source searches did not locate this exact sharp diameter theorem; that does not establish novelty. The 2017 addendum discusses asymptotic approximation, but this note does not endorse its printed quantitative formula. No novelty is claimed for the classical composition formula or elementary counterexample.

Dürer's general question remains the existence of a successful spanning-tree cut using actual edges in the original metric of every convex three-polytope. The present estimate supplies no chain-specific separation margin. An overlapping tree, an affine deformation, or the correction of a subsidiary lemma does not decide the general question.